

Capturing traffic over Wi-Fi

Auto import media

Privacy Analysis of Augmented Reality Glasses

1 Introduction

This experiment captures network traffic over Wi-Fi during the interaction between an Android phone and the Ray Ban Meta glasses, more specifically looking into the auto import of media. The aim of this data collection is to observe the network traffic generated during typical media capture and to see if there are any differences from manually importing the media onto the phone directly after capture. Auto importing the media works when the glasses are turned on and left at an idle position where the arms are closed or placed back into the charging case.

2 Metadata

Time of experiment: Oct 16th 01:10

Duration of experiment: Roughly 8 minutes

IP addresses:

- Laptop - 192.168.0.215
- Redmi phone - 192.168.1.141
- Wi-Fi Network - 192.168.0.1
- ALFA Hotspot - 192.168.1.100/24
- Meta CDN (content delivery network) - 157.240.17.17

3 Touch Commands used

- Keep idle for 1 minute (glasses off)
- Open Meta View
- Power On
- Pairing device
- Short click capture button to take a photo
- Long click capture button to take a video recording
- Short click capture button to stop video recording
- Auto-import photo/video
- Keep idle for 5 minutes (glasses on)

4 Time Stamps

5 Note

All packets: 8126

IPv4 - 97.1%

UDP - 6.3%

TCP - 90.2%

IPv4 -

Phone to 157.240.17.17 (owned by Meta CDN), 7096

Phone to 157.240.17.3 (owned by Meta), 314

Phone to 224.0.0.251 (Multicast DNS address), 152

Phone to 192.168.1.100 (ALFA Hotspot), 90

TCP -

Phone to 157.240.17.17 (owned by Meta), mostly

UDP -

Phone to 192.168.1.100 (ALFA Hotspot), roughly 90